

Geometría diferencial y aplicaciones

2º Parcial, Tarragona, 27 de mayo de 2024

Escuela Técnica Superior de Ingeniería

Nombre y apellidos:

Teoría: Enunciar 13 resultados en los cuales intervenga el teorema fundamental de la Geometría riemanniana; demostrar 4 de esos resultados.

1.- Sea (M, g) una variedad riemanniana conexa de dimensión 4. Sean Ric y G sus campos tensoriales de Ricci y de Einstein, respectivamente. Se dice que M es variedad de Einstein si $Ric = \lambda g$, con $\lambda \in \mathcal{F}(M)$ una aplicación diferenciable. Teniendo en cuenta los conceptos de divergencia y de contracción, demostrar que:

- a) M es de Einstein $\Rightarrow \lambda = \text{constante}$.
- b) $Ric = 0 \Leftrightarrow G = 0$.

2.- Sea M una variedad diferenciable de dimensión m , y $\omega \in \Omega^{k-1}(M)$ una $(k-1)$ -forma diferencial sobre M . Sean $\{X_1, \dots, X_k\} \subset \mathfrak{X}(M)$ campos diferenciables. Una de las propiedades de la diferencial exterior es que:

$$\begin{aligned} d\omega(X_1, \dots, X_k) &= \sum_{1 \leq i \leq k} (-1)^{i-1} X_i \left(\omega \left(X_1, \dots, \widehat{X_i}, \dots, X_k \right) \right) + \\ &+ \sum_{1 \leq i < j \leq k} (-1)^{i+j} \omega \left([X_i, X_j], X_1, \dots, \widehat{X_i}, \dots, \widehat{X_j}, \dots, X_k \right) \end{aligned}$$

Llamemos $D\omega(X_1, \dots, X_k)$ a la parte derecha de la fórmula y supongamos ya demostrado que es también una k -forma diferencial, $D\omega \in \Omega^k(M)$. Demostrar entonces la propiedad (o sea, demostrar la igualdad $d\omega = D\omega$) para el caso en que $\omega = f dx^1 \wedge dx^2 \wedge \dots \wedge dx^{k-1} \in \Omega^{k-1}(U)$, donde f es una función diferenciable en una carta (U, φ) de coordenadas $(x^1, \dots, x^m)$.

3.- Sea la parametrización $\vec{x}$ de $\mathbb{T}^2 \hookrightarrow \mathbb{A}^3$ en el afín euclídeo, con $R > 1$:

$$\begin{aligned} (0, 2\pi) \times (0, 2\pi) &\rightarrow \mathbb{R}^3 \\ (\theta, \varphi) &\mapsto \vec{x}(\theta, \varphi) = ((\cos \varphi + R) \cos \theta, (\cos \varphi + R) \sin \theta, \sin \varphi) \end{aligned}$$

Calcular el transporte paralelo de $\frac{\partial}{\partial \varphi} \big|_{(a+t_0, \pi+s_0)}$ hasta $(a, \pi + s_0)$, a lo largo de $\sigma(r) = (\theta(r), \varphi(r)) = (a + t_0 - r, \pi + s_0)$, $r \in [0, t_0]$, con $0 < s_0 < \varepsilon$.

$$\begin{aligned}
 \textcircled{1} \quad a) \quad \text{Ric} &= \lambda g \\
 \text{div}(\text{Ric}) &= \frac{1}{2} dR \stackrel{\downarrow \text{Ric}=\lambda g}{=} \text{div}(\lambda g) = d\lambda \\
 C_1^1(\# \text{Ric}) &= R = C_1^1(\#(\lambda g)) = \lambda C_1^1(\#g) \stackrel{\textcircled{*}}{=} \lambda \cdot 4 \Rightarrow \\
 &\quad \uparrow \text{Ric}=\lambda g \\
 &\Rightarrow dR = 4 d\lambda
 \end{aligned}
 \quad \Rightarrow$$

$$\Rightarrow dR = 4 d\lambda = 2 d\lambda \Rightarrow d\lambda = 0 \Rightarrow \lambda = \text{cte.} \quad \underline{\text{ged.}}$$

$\textcircled{*}$ Es cierta porque:

$$\begin{aligned}
 C_1^1(\#g) &= C_1^1\left(\# \left[\sum_{ij} g_{ij} dx^i \otimes dx^j \right]\right) = C_1^1\left(\sum_{ij} \left[\sum_{\alpha} g_{ij} g^{\alpha i} \right] dx^j \otimes \frac{\partial}{\partial x^i} \right) = \\
 &= C_1^1\left(\sum_{ij} \delta_j^i dx^j \otimes \frac{\partial}{\partial x^i} \right) = \sum_{\alpha=1}^4 \delta_{\alpha}^{\alpha} = 4
 \end{aligned}$$

$$b) \quad \boxed{\Rightarrow} \text{Ric} = 0 \Rightarrow R = 0 \Rightarrow \text{Ric} - \frac{R}{2} g = G = 0$$

$$\boxed{\Leftarrow} \text{Ric} - \frac{R}{2} g = G = 0 \Rightarrow \text{Ric} = \frac{R}{2} g \Rightarrow$$

$$\begin{aligned}
 \Rightarrow R &= C_1^1(\# \text{Ric}) = C_1^1\left(\# \left(\frac{R}{2} g \right)\right) \stackrel{\textcircled{*}}{=} \frac{R}{2} 4 \Rightarrow R = 0 \\
 &\quad \left. \begin{array}{l} \Rightarrow \text{Ric} = 0 \\ \underline{\text{ged.}} \end{array} \right\}
 \end{aligned}$$

② Basta demostrar que $d\omega(\frac{\partial}{\partial x^{j_1}}, \dots, \frac{\partial}{\partial x^{j_k}}) = \mathcal{L} \omega(\frac{\partial}{\partial x^{j_1}}, \dots, \frac{\partial}{\partial x^{j_k}})$
 con $j_1 < \dots < j_k$.

$$A = d\omega(\frac{\partial}{\partial x^{j_1}}, \dots, \frac{\partial}{\partial x^{j_k}}) = \sum_{i=1}^m \frac{\partial f}{\partial x^i} dx^i \wedge dx^1 \wedge \dots \wedge dx^{k-1} (\frac{\partial}{\partial x^{j_1}}, \dots, \frac{\partial}{\partial x^{j_k}}) =$$

$\omega = \int dx^1 \wedge \dots \wedge dx^{k-1}$

$$= \sum_{i=k}^m \frac{\partial f}{\partial x^i} dx^i \wedge dx^1 \wedge \dots \wedge dx^{k-1} (\frac{\partial}{\partial x^{j_1}}, \dots, \frac{\partial}{\partial x^{j_k}}) =$$

③ Índices ordenados

$$= \begin{cases} 0 & \text{si } (j_1, \dots, j_k) \neq (1, 2, \dots, k-1, \alpha) \text{ con } \alpha \geq k; \text{ o al contrario:} \\ \sum_{i=k}^m \frac{\partial f}{\partial x^i} dx^i \wedge dx^1 \wedge \dots \wedge dx^{k-1} (\frac{\partial}{\partial x^1}, \frac{\partial}{\partial x^2}, \dots, \frac{\partial}{\partial x^{k-1}}, \frac{\partial}{\partial x^\alpha}) & \text{con } \alpha \geq k \end{cases}$$

$$= \begin{cases} 0 & \text{si } (j_1, \dots, j_k) \neq (1, 2, \dots, k-1, \alpha) \text{ con } \alpha \geq k; \text{ o al contrario:} \\ \sum_{i=k}^m (-1)^{k-1} \frac{\partial f}{\partial x^i} dx^i \wedge dx^1 \wedge \dots \wedge dx^{k-1} (\frac{\partial}{\partial x^\alpha}, \frac{\partial}{\partial x^1}, \frac{\partial}{\partial x^2}, \dots, \frac{\partial}{\partial x^{k-1}}) & \end{cases}$$

$$= \sum_{i=k}^m (-1)^{k-1} \frac{\partial f}{\partial x^i} S_\alpha^i = (-1)^{k-1} \frac{\partial f}{\partial x^\alpha} \text{ con } \alpha \geq k.$$

$$B = \mathcal{L} \omega(\frac{\partial}{\partial x^{j_1}}, \dots, \frac{\partial}{\partial x^{j_k}}) = \sum_{i=1}^k (-1)^{i-1} \frac{\partial}{\partial x^{j_i}} \left(\int dx^1 \wedge \dots \wedge dx^{k-1} \right) (\frac{\partial}{\partial x^{j_1}}, \dots, \frac{\partial}{\partial x^{j_i}}, \dots, \frac{\partial}{\partial x^{j_k}})$$

$[\frac{\partial}{\partial x^{j_a}}, \frac{\partial}{\partial x^{j_b}}] = 0$

③ Índices ordenados

$$= \sum_{i=1}^k (-1)^{i-1} \frac{\partial}{\partial x^{j_i}} (0) = 0 \text{ si } (j_1, \dots, j_k) \neq (1, 2, \dots, k-1, \alpha) \text{ con } \alpha \geq k; \text{ o al contrario:}$$

$$(-1)^{k-1} \frac{\partial}{\partial x^\alpha} \left(\int dx^1 \wedge \dots \wedge dx^{k-1} \right) (\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^{k-1}}) = (-1)^{k-1} \frac{\partial f}{\partial x^\alpha} \text{ con } \alpha \geq k.$$

Porque si $i \neq k$ siempre hay un $dx^0(\frac{\partial}{\partial x^\alpha}) = 0$.

Osea $A=B$. q.e.d.

③ Paso 1: Calculo las métricas g y g^{-1} de T^2 $x^1 = \theta$
 $x^2 = \varphi$

$$\left\langle \frac{\partial \vec{x}}{\partial \theta}, \frac{\partial \vec{x}}{\partial \theta} \right\rangle = E = g_{11} = (\cos \varphi + R)^2, \left\langle \frac{\partial \vec{x}}{\partial \theta}, \frac{\partial \vec{x}}{\partial \varphi} \right\rangle = F = g_{12} = g_{21} = 0$$

$$\left\langle \frac{\partial \vec{x}}{\partial \varphi}, \frac{\partial \vec{x}}{\partial \varphi} \right\rangle = G = g_{22} = 1 \Rightarrow g^{11} = \frac{1}{(\cos \varphi + R)^2}, g^{22} = 1, g^{12} = g^{21} = 0$$

Paso 2: Calculo las Γ_{ij}^k de la conexión de Levi-Civita.

$$\Gamma_{11}^1 = 0, \Gamma_{11}^2 = \sin \varphi (\cos \varphi + R), \Gamma_{12}^1 = \Gamma_{21}^1 = -\frac{\sin \varphi}{\cos \varphi + R}, \Gamma_{21}^2 = \Gamma_{12}^2 = 0$$

$$\Gamma_{22}^1 = 0, \Gamma_{22}^2 = 0$$

Paso 3: Planteo el sistema del transporte paralelo.

$$\sigma(r) = (a + t_0 - r, \pi + s_0), \quad X(r) = \sum_i \xi^i(r) \frac{\partial}{\partial x^i} \Big|_{\sigma(r)} \Rightarrow$$

$$\Rightarrow \frac{DX}{dr} = \sum_{k=1}^2 \left(\sum_{ij} \xi^i(r) (x_0^j \sigma)'(r) \Gamma_{ji}^k(\sigma(r)) + (\xi^k(r))' \right) \frac{\partial}{\partial x^k} \Big|_{\sigma(r)} = \vec{0}$$

Teoría (def)

$$\text{Como } (x_0^1 \sigma)'(r) = (a + t_0 - r)' = -1$$

$$(x_0^2 \sigma)'(r) = (\pi + s_0)' = 0$$

$$\Rightarrow \left\{ \begin{array}{l} (k=1): \sum_i -\xi^i(r) \Gamma_{1i}^1(\sigma(r)) + (\xi^1(r))' = 0 \\ (k=2): \sum_i -\xi^i(r) \Gamma_{2i}^2(\sigma(r)) + (\xi^2(r))' = 0 \end{array} \right\} \Rightarrow \text{Paso 2}$$

$$= \left\{ \begin{array}{l} + \xi^2(r) \left(\frac{\sin \varphi}{\cos \varphi + R} \right) (\sigma(r)) + (\xi^1(r))' = 0 \\ - \xi^1(r) (\sin \varphi (\cos \varphi + R)) (\sigma(r)) + (\xi^2(r))' = 0 \end{array} \right\} \Rightarrow \left\{ \begin{array}{l} + \xi^2(r) \frac{\sin(\pi + s_0)}{\cos(\pi + s_0) + R} + (\xi^1(r))' = 0 \\ - \xi^1(r) \sin(\pi + s_0) (\cos(\pi + s_0) + R) + (\xi^2(r))' = 0 \end{array} \right.$$

Paso 4: Resuelvo el sistema.

Derivo la primera ecuación: $(\xi^1(r))'' + (\xi^2(r))' \frac{\sin(\pi+s_0)}{\cos(\pi+s_0)+R} = 0$

Substituyo la segunda ecuación en esta $\Rightarrow$
 $\Rightarrow (\xi^1(r))'' + \xi^1(r) \sin^2(\pi+s_0) = 0$

$\Rightarrow \xi^1(r) = K_1 \cos(\sin(\pi+s_0)r) + K_2 \sin(\sin(\pi+s_0)r)$
Resuelvo (lineal homogénea) $\Downarrow$

$$\xi^2(r) = -(\xi^1(r))' \frac{\cos(\pi+s_0)+R}{\sin(\pi+s_0)} =$$

$$= K_1 \sin(\sin(\pi+s_0)r) (\cos(\pi+s_0)+R) - K_2 \cos(\sin(\pi+s_0)r) (\cos(\pi+s_0)+R)$$

Paso 5: Impongo las condiciones iniciales.

$$\underline{X}(0) = \frac{\partial}{\partial \varphi} \Big|_{(a+t_0, \pi+s_0)} = \frac{\partial}{\partial x^2} \Big|_{(a+t_0, \pi+s_0)} \stackrel{r=0}{=} K_1 \frac{\partial}{\partial x^1} + (-K_2 (\cos(\pi+s_0)+R)) \frac{\partial}{\partial x^2} \Big|_{(a+t_0, \pi+s_0)} \Rightarrow$$

$\uparrow$ $\xi^1(0)$; $\xi^2(0)$

$$\Rightarrow \begin{cases} K_1 = 0 \\ -K_2 (\cos(\pi+s_0)+R) = 1 \end{cases} \Rightarrow$$

$$\Rightarrow \underline{X}(r) = -\frac{\sin(\sin(\pi+s_0)r)}{\cos(\pi+s_0)+R} \frac{\partial}{\partial x^1} (\sigma(r)) + \cos(\sin(\pi+s_0)r) \frac{\partial}{\partial x^2} (\sigma(r))$$

Paso final: Substituyo en $\underline{X}(r)$, $r = t_0$

$$\int_{0, \sigma}^{t_0} \left(\frac{\partial}{\partial \varphi} \Big|_{(a+t_0, \pi+s_0)} \right) = \frac{-\sin(t_0 \sin(\pi+s_0))}{\cos(\pi+s_0)+R} \frac{\partial}{\partial \theta} \Big|_{(a, \pi+s_0)} + \cos(t_0 \sin(\pi+s_0)) \frac{\partial}{\partial \varphi} \Big|_{(a, \pi+s_0)}$$